

Electric Vehicle Charging Infrastructure in India: Viability Analysis

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Abstract Rising pollution is a cause of major concern for the Indian cities. Extensive reliance on IC engine-based vehicles as the principal means of transport has raised serious environmental concerns. To address the ongoing problems, there is increased emphasis on developing high-efficient, emission-free means of transport. Keeping this in view, electric and hybrid vehicles appear to be the best alternatives for replacing the conventional vehicles. But wide adoption of electric vehicles will require residential as well as public charging infrastructure analogues to petrol pumps. Level 1, Level 2, and fast EV charging stations have been installed in various countries in the world. A detailed exploration is required to develop a similar customized infrastructure for the Indian market. This work provides a detailed overview of the various technologies and viable options in terms of power quality, power electronic converter topologies, and energy management options optimally feasible in India. Recent policies and initiatives taken by the Government are also presented in this paper.

Keywords Charging infrastructure • Electric vehicles • Smart grid
Smart charging management

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1 Introduction

The ongoing smart city endeavor in India has to address challenges including air pollution, increasing greenhouse gas emissions, and rising risks of energy security. Many cities in India have extremely high levels of urban air pollution in the form of oxides of carbon, sulfur, and nitrogen [1].

Transport sector is the major contributor to the environmental pollution accounting for about 51% of pollution of India as shown in Fig. 1, and in urban areas, this figure goes to 75–80%. There is growing interest in alternative fuel vehicles, particularly in electric vehicles (EVs) to minimize the negative impact of transport sector on environment. EVs have not gained wide acceptance among customers in the past. However, technological advancements, particularly in the field of battery technology, have made EVs attractive to the customers. EVs have started penetrating the automotive market in many Indian cities. However, large-scale deployment of EVs in India largely depends on the availability of EV charging infrastructure which includes slow charging stations (Level 1/Level 2) for home charging and fast charging stations situated in public places for commercial charging.

2 Building Blocks of EV Charging Infrastructure

EV charger is an integral part of the charging infrastructure. Charging time of EV is closely associated with the charger characteristics. Charger should be efficient, with high power density, low weight, and low volume. An EV charger must draw utility current with low distortions so that there is minimum impact on the power quality of grid. With conventional rectifiers, a large proportion of low-order harmonics are injected in the source current due to which power factor decreases. Poor power factor leads to inefficient utilization of volt-ampere rating. To mitigate these adverse effects, power factor correction circuits are used with the converters. Numerous

Fig. 1 Air pollution caused by different sectors in India [2]

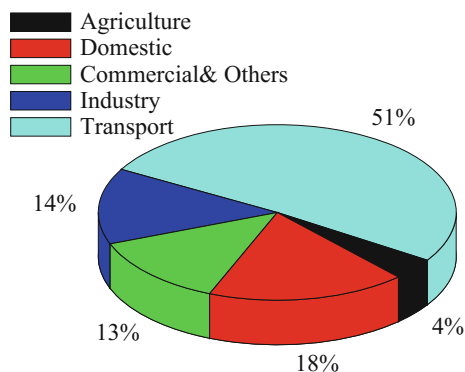


Table 1 EV converter topologies [4]

Topology	Conventional boost	Phase-shifted semi-bridgeless	Interleaved	Bridgeless interleaved
Power capacity	<1 kW	<3.5 kW	<3.5 kW	>5 kW
Output ripples	High	Medium	Low	Low
Input ripples	High	Medium	Low	Low
Efficiency	Poor	Best	Fair	Best
Cost	Low	Medium	Medium	High

control techniques and circuit topologies have been developed to achieve power factor correction. Table 1 gives the comparison between different converter topologies on the basis of power rating, capacitor ripples, input distortion, efficiency, and cost.

2.1 Power Levels of Charging

On the basis of charging power levels, EV charging can be classified as Level 1 charging, Level 2 charging, Level 3 charging, and DC fast charging.

Level 1 charging is the slowest mode of charging which requires 120 V/15 A [3], single-phase supply and J1772 EV connector to connect with the EV port. It usually takes place at home or garage. Level 2 charging is used for both public and private applications. It uses dedicated supply equipment which is connected to 208 or 240 V outlet. Level 3 and fast DC charging are used for commercial applications which take place at public places like highways, city fueling points, similar to gas stations.

Table 2 EV charging levels

Charging type	Level 1	Level 2	DC fast
Charging time (h)	20–22	6–8	0.2–0.5
Charger location	Onboard (1-phase)	Onboard (1 or 3-phase)	Off-board (3-phase)
Voltage supply (V)	120	240	208–600
Power level (kW)	1.3–1.9	Up to 19.2	50–150
Range	2–5 miles per hour of charging	10–20 miles per hour of charging	60–80 miles in <30 min
Primary use	Residential charging	Residential and public charging	Public charging

Table 2 gives the description of various charging levels based on charging time, power and voltage levels, range obtained in per hour of charging, and charger location.

2.2 Standards of EV Charging

Organizations such as International Electrotechnical Commission (IEC), Society of Automotive Engineers (SAE), Institute of Electrical and Electronics Engineers Standards Association (IEEE-SA), Japan Electric Vehicle Standards (JEVS), Infrastructure Working Council (IWC) have prepared codes and standards for development of EV charging infrastructure. Some of the standards are listed in Table 3.

2.3 Onboard and Off-board EV Charging Topologies

EV chargers can be classified as onboard and off-board chargers, with unidirectional and bidirectional power flow capabilities. A charger with bidirectional capability allows charging from the grid as well as energy injection back to the grid.

Onboard chargers are restricted by size and weight limitation due to limited space on the vehicle, while there is no such limitation for off-board chargers.

Table 3 EV charging standards [6]

Standard	Specification
SAE-J1772	EV coupler for conductive charging
SAE-J1773	EV inductively coupled charging
SAE-J1797	Recommended practice for EV battery modules packaging
SAE-J2288	Life cycle testing of battery modules for EV
SAE-J2464	EV/HEV rechargeable energy storage system (RESS) safety and abuse testing
SAE-J2836 Part 1	Use cases for communications between PEVs and utility grid
SAE-J2836 Part 2	Use cases for communications between PEVs and supply equipment (EVSE)
SAE-J2836 Part 3	Use cases for communications between plug-in vehicles and the utility grid for reverse flow
SAE-J2894	Power quality requirements for plug-in vehicle chargers—requirements
IEC-69/156/CD:2008	Electric vehicle conductive charging system
IEC-23H/222/CD:2010	Plugs, socket outlets, vehicle couplers, and vehicle inlets—conductive charging of EVs
JEVS-C601:2000	Plugs and receptacles for EV charging
AIS-138 (Draft)	Electric vehicle conductive AC charging system-ARAI

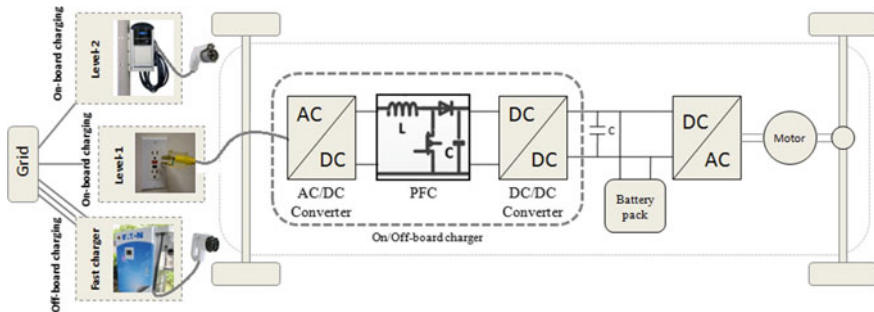


Fig. 2 Typical layout of EV charging system with on/off-board charging strategies

Off-board chargers are used for fast charging, while Level 1 charging and Level 2 charging schemes extensively use onboard chargers [5].

Figure 2 shows a typical layout of EV charger with onboard and off-board charging schemes for different levels of charging. In case of fast charging, the charger assembly consisting of AC/DC and DC/DC converters is shifted off-board and rectified output of the converter is given to the EV charging inlet.

3 Recommended Planning of EV Charging Infrastructure

Market penetration of EVs is strongly dependent on the charging infrastructure availability. Hence, development of infrastructure for EV charging is considered as the foremost task by concerned governments and electric vehicle developers. An important factor which is to be considered in this regard is that the existing power system must be capable of accommodating this new charging load. For that, an approach for charging that is consumer convenient and at the same time does not degrade the power grid is required.

Availability of commercialized charging stations would offer certain benefits to the power grid. Some of them are:

- They will help in diverting the peak of charging load from the demand peak of the network.
- Unpredictable mobile load in the form of EVs would be transformed into a stationary load, and it would be easier to predict.
- When in the form of bulk charging load, it would be simpler to enforce regulations on harmonics and power factor.
- Implementation of V2G concept would be easy as it would eliminate the need for integration of sophisticated devices for measurement, communication, and control, up to end consumer level.

Moreover, when compared with slow Level 1 and Level 2 charging modes, which take considerable time, fast charging at public charging stations would definitely be a preferable choice for EV consumers too. Hence if prior actions are taken by utilities for the development of charging infrastructure well before in time, it would prove to be of great benefit. For that, appropriate planning is important, incorporating identification of suitable sites and charging load size, satisfying the required constraints for planning.

A more realistic approach for planning would be to select locations for charging stations which satisfy the requirements of both consumers and power system. From that perspective, the location which satisfies the given criteria is considered to be ideal.

- The location should be in the vicinity of EV consumers.
- The location should be so chosen that it covers the productive EV sites like town centers, office complexes, residential areas, road networks.
- Location of charging station should be such that the regulatory limits on voltage and on line MVA flows are not violated and least losses are incurred in the power system.

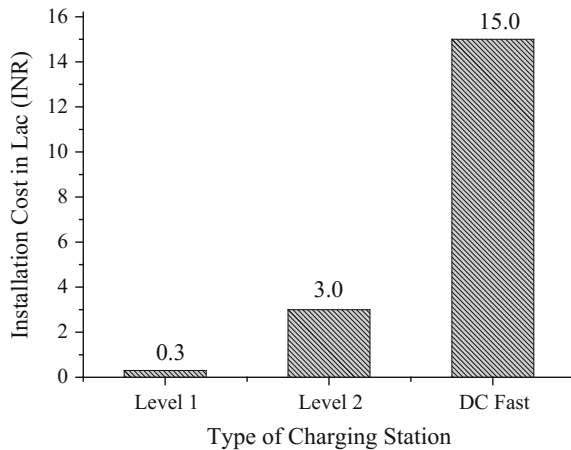
Numbers of optimization techniques like genetic algorithm, simulated annealing, virtual ant algorithms, particle swarm optimization are used for the planning of charging infrastructure. Particle swarm optimization is used in [7] using an IEEE 16 bus distribution test system.

4 Cost Estimation of Charging Infrastructure

Total investment cost required for the establishment of charging infrastructure for EVs includes the cost of equipments to be used, installation costs, and operation and maintenance costs. With increase in penetration of EVs in the next few years, number of EV chargers will increase and hence the equipment cost is expected to decrease. There is large variation in the EV charging equipment cost among different manufacturers. Besides the initial equipment cost, installation cost is required for installation of the charger and interconnection with grid. The installation cost includes cost of civil works, transaction cost regarding distribution system operator permission, and other related costs depending on factors like requirement of a new grid connection or upgradation of the existing connection. In case there is a preexisting connection, the installation cost may greatly reduce. Some other parameters that influence installation cost include number of simultaneous installation of many chargers which may reduce cost on account of common labor and grounding costs, mutual components, etc.

For semiprivate/semipublic places where low- or medium-power level chargers are required, cost varies between 500 € (Rs. 36,431) and 1200 € (Rs. 87,435) [8, 9]. For public places where high power level chargers are required, installation cost is relatively higher ranging between 2400 € (Rs. 1,74,871) to 3600 € (Rs. 2,62,306) [9]. Apart from the equipment and installation costs, there is a requirement for

Fig. 3 Estimated installation cost of EV charging stations



continuous maintenance over the running period. For that, operation and maintenance cost is to be added, which may be taken to be 10% of the total installation cost (including equipment cost) [10].

Figure 3 shows the cost of installing EV charging stations in Delhi as estimated in [11] for Level 1 charger of 1.5 kW, Level 2 charger of 6.6 kW, and DC fast charger of 50 kW capacity.

5 Impact on Power Grid

There is large randomness in the behavior of EV charging, which is greatly influenced by factors like mileage, battery capacity, charging time. Interconnection of charging infrastructure with the grid has some effects on the distribution network which are discussed here.

- The first issue that comes on interconnecting charging infrastructure is that it increases the difficulty of distribution network planning. New constraints in the form of electricity demand and the layout of charging stations need to be considered, which add to the complexity of network planning.
- Improved quality equipments with high ratings are required in the distribution network to facilitate interconnection of charging infrastructure. Integration of high-power charging station requires distribution transformer with larger capacity and distribution line of larger cross section to avoid problems like overloading, voltage deviation.
- It may lead to a decrease in the economy of distribution system operation. As charging load exhibits large volatility, it is difficult to confine charging behavior to low-load periods, leading to greater system peak difference. This would ultimately result in lower utilization efficiency of distribution network equipment.

- Power quality of the distribution network is affected. Charger uses several power electronic conversion devices in the form of converters which induce harmonics in the source-side current.

6 Need for Smart Charging Management

With increasing EV penetration in the Indian scenario, the demand for charging is expected to increase proportionally, and it would act as an additional burden on the grid. This additional load is bound to affect the grid adversely, if optimal scheduling is not done in advance. It will lead to imbalance of voltage as well as frequency, which ultimately may lead to grid failure and blackout. Continuous monitoring of power system is necessary while charging large number of EVs, in order to achieve grid balancing. The concept of smart grid, wherein a two-way communication can be established between the utility and consumers, has to be reinforced. A typical demand profile for one day, obtained from Power Exchange (IEX), is shown in Fig. 4 which is a replica of the energy demand in Indian scenario, where minimum load is observed during the night hours and also during mid of the day.

Low peak hours during night, wherein the load is at its minimum, are most suitable for home-based charging, using either Level 1 or Level 2. This period of 6–8 h is also very good for charging batteries at battery swapping stations (BSS). Public charging stations (CSs), where most of the charging is expected to occur during daytime, can shift their charging load to low peak hours to enhance the functioning of grid. This can be done by deploying attractive charging schemes for the consumers during low peak hours or by storing energy during this period as a backup for peak load hours. This type of scheduling can prove to be beneficial for both the utility and consumers. A flatter load profile with less system peak differences can be obtained this way, which is desirable. Also, from the consumer's

Fig. 4 Demand profile for one day

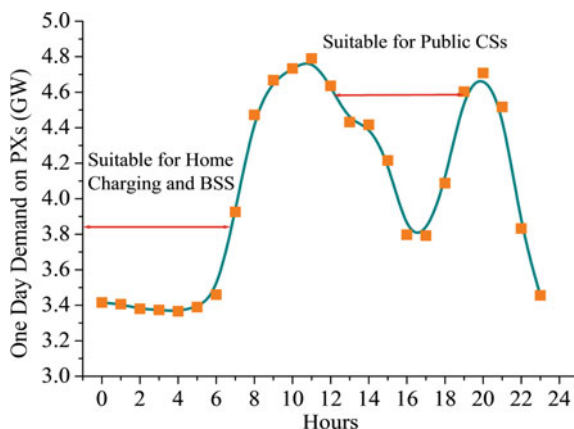
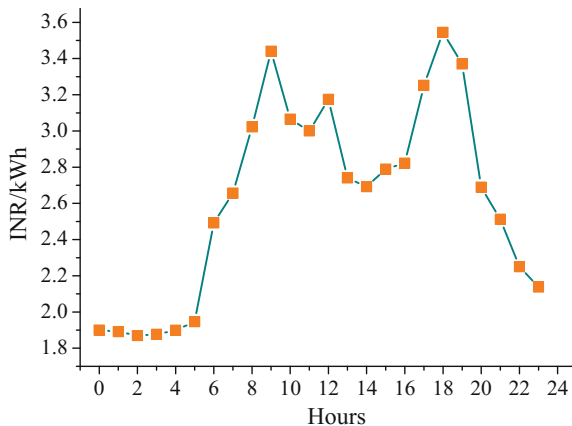


Fig. 5 Cost of per unit electricity



viewpoint, this is advantageous as evident from Fig. 5. Cost of purchasing electricity is lower during low peak hours as compared to that during peak hours.

Such a demand-side management can help in smooth running of power grid with fewer disturbances. Further enhancement of grid operation can be achieved by injection of power back to the grid using vehicle to grid (V2G) concept and also by participating in reserve market to maintain the frequency up and down regulations. This can prove to be beneficial for the EV aggregator as well as to consumers, in terms of economy.

The aggregator of EV charging station as well as BSS can gain maximum profit by optimizing the cost function of trading revenue mentioned in Eq. (1). An optimization model based on this cost function is given in [12].

$$\max\{r^{\text{em}} + r^{\text{cap}} - c^{\text{regup}} - c^{\text{regdn}}\} \quad (1)$$

where r^{em} represents the day-ahead (DA) revenue of energy market, r^{cap} represents the DA regulation market revenue for the capacity, c^{regup} represents regulation up service cost, and c^{regdn} represents regulation down service cost.

The DA revenue of energy market r^{em} can be expressed as:

$$r^{\text{em}} = \Delta t \sum_{t \in T} \sum_{v \in V} \lambda_t^{\text{DA}} (\eta_v^{\text{dsg}} \cdot p_{t,v}^{\text{emdsg}} - p_{t,v}^{\text{emchg}}) \quad (2)$$

where λ_t^{DA} is DA energy market price, η_v^{dsg} is battery discharge efficiency, $p_{t,v}^{\text{emdsg}}$ and $p_{t,v}^{\text{emchg}}$ represent discharging and charging powers, respectively.

The revenue r^{cap} is obtained as:

$$r^{\text{cap}} = \sum_{t \in T} \sum_{b \in B} [(w_{t,b}^{\text{up}} \lambda_{t,b}^{\text{up}}) \pi^a p_t^{\text{up}} + (w_{t,b}^{\text{dn}} \lambda_{t,b}^{\text{dn}}) \phi^a p_t^{\text{dn}}] \quad (3)$$

where $\lambda_{t,b}^{\text{up}}$ and $\lambda_{t,b}^{\text{dn}}$ are DA regulation up and down capacity prices, respectively; power p_t^{up} and p_t^{dn} are regulation up and down capacities offer to the market; $\pi^{a/d}$ is the probability of acceptance and deployment, respectively, for up regulation; $\phi^{a/d}$ is the probability of acceptance and deployment, respectively, for down regulation; $w_{t,b}^{\text{up/dn}}$ and $v_{t,b}^{\text{up/dn}}$ are the segment activation of the price quantity for capacity and real-time energy probability curves in period t and segment b .

Cost for regulation up service (c^{regup}) can be expressed as:

$$c^{\text{regup}} = \pi^a \pi^d (1 - \pi^d) \sum_{t \in T} \sum_{b \in B} (v_{t,b}^{\text{up}} \lambda_{t,b}^{\text{RT}}) (p_t^{\text{up}} - \pi^a \pi^d p_t^{\text{up}}) \quad (4)$$

Similarly, cost for regulation down service is expressed as:

$$c^{\text{regdn}} = \Phi^a \Phi^d (1 - \Phi^d) \sum_{t \in T} \sum_{b \in B} (v_{t,b}^{\text{dn}} \lambda_{t,b}^{\text{RT}}) (p_t^{\text{dn}} - \Phi^a \Phi^d p_t^{\text{dn}}) \quad (5)$$

In order to participate in DA market, proper load forecasting is essential. Numerous techniques have been developed for load forecasting like simplistic benchmark methods, seasonal ARMA modeling, periodic AR models [13]. For simplistic benchmark method, the forecast function is given as:

$$\overline{y_{(t)}}(k) = y_{t+k-s_2} \quad (6)$$

where y_t represents the demand in period t and k is the forecast lead time ($k \leq s_2$). Second simplistic benchmark does simple averaging of the corresponding observations in the past for forecasting. The forecast function for this method is given in Eq. (7) for four observations.

$$\overline{y_{(t)}}(k) = (y_{t+k-s_2} + y_{t+k-2s_2} + y_{t+k-3s_2} + y_{t+k-4s_2})/4 \quad (7)$$

The uncertainties involved in the EV fleet characteristics, DA electricity market operations as well as in generation, transmission, and load may be represented by Monte Carlo simulation. Risk involved with the financial as well as economical aspects of the EV aggregator in uncertainty environments can be managed by conditional value-added risk analysis (CVaR) as given in Eq. (8).

$$\text{Maximize}_{\zeta, \delta^s} \text{CVaR} = \zeta - \frac{1}{(1-a)} \sum_{s=1}^{\text{NS}} \text{Pr}^s \cdot \delta^s \quad (8)$$

For detailed risk analysis based on the above method, readers may go through [14].

7 Key Barriers and Potential Solutions

Scaling up EV penetration in Indian automotive market requires addressing certain challenges and barriers related to infrastructure, market, technical limitations, and policies. Technical challenges include dealing with issues like improvement in battery technology, reduction in charging time, and getting sufficient driving range for EVs. Low specific energy density of EV batteries like lead–acid batteries is a major barrier. So to achieve higher driving ranges, bulky batteries are required, which increase the overall weight of the vehicle. To overcome this issue, battery makers are working on developing batteries having high values of specific energy density. One major concern is that India has lack of lithium-ion reserves, which give high energy density batteries [15].

Market and infrastructural barriers are related to lack of charging infrastructure, absence of business models to meet specific needs of EVs, and lack of dedicated lanes. Moreover, commercial stations for charging and battery swapping should be there at public places to allow ease of charging. Similar to technical barriers, solutions exist for market and infrastructural barriers also. To address the problem of charging infrastructure, a solution could be to set up charging points in parking garages and in the basements of buildings. This strategy has been adopted by many countries like China. There is also a need to build business models which may reduce upfront investment of consumers and increase consumer confidence in electric vehicles.

Lack of well-defined policy framework for the promotion of EVs in the country is a major barrier to widespread adoption of EVs. Policymakers need to implement a policy road map keeping in view the benefits provided by EVs. Governments need to support research activities in battery technology and vehicle innovations, along with providing incentives to enhance EVs production and sale. There is also a need to understand the public perceptions regarding safety, driving range, and cost of these vehicles compared to conventionally powered vehicles and make policies in light of them. National policies like NEMMP may prove to be a major step toward this goal.

8 EV Market in India

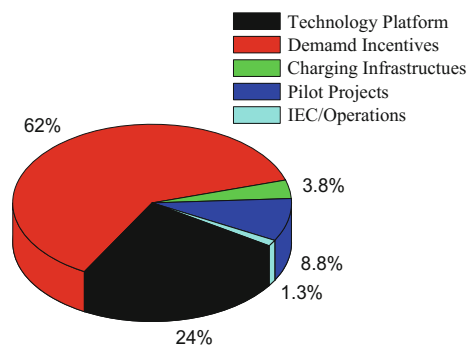
The present EV market in India is not so large. Though there are plenty of electric two-wheelers and four-wheelers in the market, but even then EVs' overall share is negligible. Electric vehicles started penetrating the Indian automotive market long back in the 1990s, when electric two- and three-wheelers were first launched. But they did not prove to be successful and were later discontinued. A major breakthrough came in the early 2000s with the launch of first electric car in India by Mahindra Electric, named Reva. Mahindra Electric emphasized on developing electric cars at reasonable prices. It launched its first model in India in the year 2001

[16]. A modified version of this car came in the market in the year 2013. This new model used lithium-ion powered batteries and allowed for driving range of 100 km with full charge. Prius hybrid model was launched by Toyota in Indian market in the year 2010. This was followed by Camry Hybrid in 2013. In few Indian cities, hybrid and electric buses have been introduced. Municipal Transport Corporation of Bangalore launched an electric bus, few years back. Electric rickshaws have emerged as excellent substitutes for the conventional three-wheeled vehicles as well as for the paddle rickshaws. But still there is vast scope for the expansion of EV market in the country.

9 Policy Framework in India

Several initiatives have been taken by the government–industry collaboration to promote the widespread adoption of EVs in the country. “Ministry of New and Renewable Energy (MNRE)” has promoted the R&D work on electric vehicles as part of the Alternative Fuel for Surface Transportation program. “National Electric Mobility Mission Plan (NEMMP) 2020” [17] has been launched by Department of Heavy Industry for the promotion of EVs in India. The mission aimed at deploying 6–7 million electric vehicles in India by the year 2020. A scheme has been formulated named “FAME India (Faster Adoption and Manufacturing of (Hybrid and) Electric Vehicles in India)” [18], as a part of this mission. This scheme is to be implemented over 6-year period, till year 2020. This scheme focuses on four major areas including technological development, charging infrastructure, pilot projects, and demand creation. The percentage breakup of total fund (Rs. 795 crore) allocated to key areas is given in Fig. 6.

Fig. 6 Fund allocation under FAME India scheme [19]



10 Conclusion

India ought to put resources into little scale fortifications to deal with the electric load issues locally, rather than going for an enormous change. Home charging ought to be keyed up for long battery life and framework adjustment. Legitimate arrangement of place, populace, traffic density behavior, and the security ought to be thought about before going for a large-scale charging infrastructure execution for the second biggest populated nation of the world. The most essential aspect in the working of an electric vehicle charging framework is the incorporation of exercises in the electrical vitality and transport fields. Just deliberate improvement of both frameworks will give steady and dependable electrical power framework, particularly at the level of restricted power and feasible advancement of the Indian electric vehicle advertise. Development of V2G concept along with G2V implementation on large scale would help in dealing with grid-related issues.

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